

Python Automata Library (PAL) Cheat Sheet

\1 · Safe mode is default. Call **\1** before constructing any PAL object. Put substantive model kernels under **\1**.

Create state

\1 — typed lattice

\1 — individual agents; **\1** is nonspatial

\1 — integer population counts

\1 — continuous field

\1 · **\1** — integer query list · random-count sampler

Dimensions — 1–3 axes; a negative dimension wraps that axis, e.g. **\1**.

Shared lattice geometry

Properties — **\1**, **\1**, **\1**

Index conversion — **\1**; **\1**

Regions — **\1** uses half-open bounds; **\1** maps relative offsets to coordinates.

Neighborhoods — **\1** · **\1** · **\1**

Grid / common indexing

Read/write — **\1**, **\1**; slices return detached NumPy copies.

Pattern — **\1** · **\1** · **\1**

AgentGrid

Create / move — **\1** / **\1** for lattice positions; **\1** / **\1** for continuous positions.

Agent state — **\1** property; **\1**, **\1** lattice; **\1** continuous; **\1**, **\1**.

Queries — **\1**, **\1**, **\1**, **\1**, **\1**.

Iteration — **\1** is a snapshot and is safe for structural mutation such as **\1**.

Wrapping — **\1** gives wrapped displacement.

PopGrid + PDEgrid

Transactional update — **\1** changes pending state; **\1** applies it simultaneously; **\1** clears current + pending. Direct **\1** changes current state immediately.

Population — **\1** total; **\1** nonzero site indices. **\1** limits total population.

PDE setup — **\1**

Diffusion — **\1**, **\1**, **\1**, **\1**, **\1**; radial 1D: **\1**.

Advection — **\1**, **\1**, **\1**.

Lists + randomness

IList — **\1**, **\1**, **\1**, **\1**, indexing/**\1**; **\1** detached copy; **\1** no-copy iteration.

RNG — **\1** · **\1** · **\1** → **\1**; use PAL RNG for shared-stream reproducibility.

Multinomial — **\1**; **\1** then **\1** for repeated multinomial draws.

Draw / output

Pixels — **\1**; set **\1**; **\1**; **\1**; **\1**.

GIF — **\1** · **\1** · **\1**; call **\1** before capture.

OpenGL — **\1**; draw with **\1**, **\1**, **\1**, **\1**, **\1**; scene controls include **\1**, **\1**, **\1**.

More detail: [Manual](#) · [API Guide](#) · [API Reference \(Documentation\)](#).